

HE SUN

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Research Interests computational imaging, adaptive optics, machine learning, and inverse problems

Education

09/2014-09/2019	Ph.D.	Mechanical and Aerospace Engineering <i>Thesis: "Efficient wavefront sensing and control for space-based high-contrast imaging"</i> <i>Advisor: Prof. N. Jeremy Kasdin</i>	Princeton University
09/2011-07/2014	B.Sc (Double Majors)	Engineering Mechanics & Economics	Peking University

Employment

05/2022-present	Assistant Professor of Biomedical Engineering, Peking University <i>Principal Investigator of Computational Scientific Imaging Lab (https://ai4imaging.github.io/)</i>		
10/2019-03/2022	Postdoctoral Scholar at Computing and Mathematical Sciences, California Institute of Technology <i>Advisor: Prof. Katherine L. Bouman. Part of the Event Horizon Telescope Collaboration.</i>		
06/2018-09/2018	Research Intern, Mitsubishi Electric Research Laboratories - Cambridge, MA <i>Host: Dr. Yebin Wang and Dr. Dong Tian</i>		

Honors & Awards

2021	Best Paper Award, Machine Learning for Health Conference
2020	Amazon AI4Science Fellowship, California Institute of Technology
2019	Best Paper Award for Observation Systems and Technologies, IEEE Aerospace Conference, Big Sky
2018	Britt & Eli Harari Fellowship, Princeton University
2017	School of Engineering and Applied Science Award for Excellence, Princeton University. "This award is given to SEAS advanced graduate students who have performed at the highest level as scholars and researchers."
2014	Outstanding Graduate Award, Peking University and Beijing Government (Top 5%)
2013	ExxoMobil Scholarship, Peking University
2012	Boeing Scholarship, Peking University
2011	1987 Alumni Scholarship, Peking University

Publications

Peer-reviewed Conference Papers

1. Gao, Angela F, Oscar Leong, **He Sun**, and Katherine L Bouman (2023). Image Reconstruction without Explicit Priors. In: *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, pp.1–5.
2. Ye, Enze, Yuhang Wang, Hong Zhang, Yiqin Gao, Huan Wang, and **He Sun** (2023). Recovering a Molecule's 3D Dynamics from Liquid-phase Electron Microscopy Movies. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp.10767–10777.
3. **Sun, He** and Katherine Bouman (2021). Deep probabilistic imaging: uncertainty quantification and multi-modal solution characterization for computational imaging. In: *35th AAAI Conference on Artificial Intelligence*.
4. Yin, Tianwei, Zihui Wu, **He Sun**, Adrian Dalca, Yisong Yue, and Katherine Bouman (2021). End-to-end sequential sampling and reconstruction for MR Imaging. *Machine Learning for Health Conference*. **Best paper**.
5. **Sun, He**, Adrian Dalca, and Katherine Bouman (2020). Learn a probabilistic strategy for computational imaging sensor selection. In: *International Conference on Computational Photography*. IEEE.
6. **Sun, He**, Jessica Gersh-Range, and N Jeremy Kasdin (2019). Modern wavefront control for space-based exoplanet coronagraph imaging. In: *Aerospace Conference*. IEEE. **Best paper for observation systems and technologies**.
7. Takahashi, Takeshi, **He Sun**, Dong Tian, and Yebin Wang (2019). Learning heuristic functions for mobile robot path planning using deep neural networks. In: *29th International Conference on Automated Planning and Scheduling*.

Journal Papers

1. Cheng, Bin, Enze Ye, **He Sun**, and Huan Wang (2023). Deep learning-assisted analysis of single molecule dynamics from liquid-phase electron microscopy. *Chemical Communications* **59**(12), 1701–1704.
2. Fuentes, Antonio, José L Gómez, José M Martí, Manel Perucho, Guang-Yao Zhao, Rocco Lico, Andrei P Lobanov, Gabriele Bruni, Yuri Y Kovalev, Andrew Chael, Kazunori Akiyama, Katherine L Bouman, **He Sun**, et al. (2023). Filamentary structures as the origin of blazar jet radio variability. *Nature Astronomy*, 1–9.
3. Leong, Oscar, Angela F Gao, **He Sun**, and Katherine L Bouman (2023). Discovering Structure From Corruption for Unsupervised Image Reconstruction. *IEEE Transactions on Computational Imaging* **9**, 992–1005.
4. **Sun, He**, Paul Tiede, Jason J. Wang, Sarah Blunt, Dimitri Mawet, and Katherine Bouman (2022). α -Deep Probabilistic Inference (α -DPI): efficient uncertainty quantification from exoplanet astrometry to black hole feature extraction. *Astrophysical Journal (ApJ)*.

5. The Event Horizon Telescope Collaboration, et. al (2022a). First Sagittarius A* Event Horizon Telescope results. I. The shadow of the supermassive black hole in the center of the Milky Way. *The Astrophysical Journal Letters* **930**(2), L12.
6. — (2022b). First Sagittarius A* Event Horizon Telescope Results. II. EHT and Multi-wavelength Observations, Data Processing, and Calibration. *The Astrophysical Journal Letters* **930**(2), L13.
7. — (2022c). First Sagittarius A* event horizon telescope results. III. Imaging of the galactic center supermassive black hole. *The Astrophysical Journal Letters* **930**(2), L14.
8. — (2022d). First Sagittarius A* Event Horizon Telescope results. IV. Variability, morphology, and black hole mass. *The Astrophysical Journal Letters* **930**(2), L15.
9. — (2022e). First Sagittarius A* event horizon telescope results. V. Testing astrophysical models of the galactic center black hole. *The Astrophysical Journal Letters* **930**(2), L16.
10. — (2022f). First Sagittarius A* event horizon telescope results. VI. Testing the black hole metric. *The Astrophysical Journal Letters* **930**(2), L17.
11. Haffert, Sebastiaan Y., Jared R. Males, Laird M. Close, Kyle van Gorkom, Joseph D. Long, Alex Hedglen, Olivier Guyon, Lauren Schatz, Maggie Kautz, Jennifer Lumbres, Rodack Alex, Knight Justin M., He Sun, and Kevin Fogarty (2021). Data-driven subspace predictive control of adaptive optics for high-contrast imaging. *Journal of Astronomical Telescopes, Instruments, and Systems*.
12. Hu, Mengya, Anthony Harness, He Sun, and N Jeremy Kasdin (2021). Exoplanet detection in starshade images. *Journal of Astronomical Telescopes, Instruments, and Systems*.
13. Hu, Mengya, He Sun, Anthony Harness, and N Jeremy Kasdin (2021). Bernoulli generalized likelihood ratio test for signal detection from photon counting images. *Journal of Astronomical Telescopes, Instruments, and Systems*.
14. Llop-Sayson, Jorge et al. (2021). Constraining the Orbit and Mass of epsilon Eridani b with Radial Velocities, Hipparcos IAD-Gaia DR2 Astrometry, and Multiepoch Vortex Coronagraphy Upper Limits. *The Astronomical Journal* **162**(5), 181.
15. The Event Horizon Telescope Collaboration, et. al. (2021a). First M87 Event Horizon Telescope Results VII: Polarization of The Ring. *Astrophysical Journal Letters*.
16. The Event Horizon Telescope Collaboration, et. al (2021b). First M87 Event Horizon Telescope Results VIII: Magnetic Field Structure Near The Event Horizon. *Astrophysical Journal Letters*.
17. Sun, He, Alexei Goun, Susan Redmond, Michael Galvin, Tyler Groff, and N Jeremy Kasdin (2020). High-Contrast Integral Field Spectrograph (HCIFS): multi-spectral wavefront control and reduced-dimensional system identification. *Optics Express*.
18. Sun, He, N. Jeremy Kasdin, and Robert Vanderbei (2020). Efficient wavefront sensing for space-based adaptive optics. *Journal of Astronomical Telescopes, Instruments, and Systems*.
19. Sun, He, N Jeremy Kasdin, and Robert Vanderbei (2018). Identification and adaptive control of a high-contrast focal plane wavefront correction system. *Journal of Astronomical Telescopes, Instruments, and Systems*.

Other Conference Proceedings

1. Haffert, Sebastiaan Y., Jared Males, Laird Close, Kyle van Gorkom, Joseph Long, Alexander Hedglen, Lauren Schatz, Jennifer Lumbres, Alexander Rodack, Justin Knight, et al. (2021). Data-driven subspace predictive control: lab and on-sky demonstration. In: *Techniques and Instrumentation for Detection of Exoplanets X*. Vol. 11823. International Society for Optics and Photonics, pp.118231C.
2. Hu, Mengya, He Sun, and N Jeremy Kasdin (2019). Sequential generalized likelihood ratio test for planet detection with photon-counting mode. In: *Proceedings of SPIE*.
3. Delacroix, Christian, He Sun, Michael Galvin, Mary Anne Limbach, Tyler Groff, Maxime Rizzo, Matthew Grossman, Katherine Mumm, and N Jeremy Kasdin (2018). First light of the High Contrast Integral Field Spectrograph (HCIFS). In: *Proceedings of SPIE*.
4. Sun, He and N Jeremy Kasdin (2018). Neural network control of the high-contrast imaging system. In: *Proceedings of SPIE*.
5. Sun, He, N Jeremy Kasdin, and Robert Vanderbei (2017). Identification of the focal plane wavefront control system using EM algorithm. In: *Proceedings of SPIE*.
6. Sun, He, N Jeremy Kasdin, Robert Vanderbei, AJ Eldorado Riggs, and Tyler Groff (2017). Improved high-contrast wavefront controllers for exoplanet coronagraphic imaging systems. In: *Proceedings of SPIE*.
7. Sun, He, AJ Eldorado Riggs, N Jeremy Kasdin, Robert Vanderbei, and Tyler Groff (2016). Modified deformable mirror stroke minimization control for direct imaging of exoplanets. In: *Proceedings of SPIE*.

Invited Talks

1. "AI accelerated ultrasound computed tomography reconstruction." *China Medicinal Biotechnology Association (CMBA) Biomedical Imaging Science Retreat*. Beijing, China. November 2023.
2. "Deep Learning Based Full Waveform Inversion for Optical Diffraction Tomography." *International Congress on Ultrasonics (ICU)*. Beijing, China. September 2023.
3. "Computational Biomedical Imaging with AI + Physics Modeling."
 - (a) *The 2nd International Computational Imaging Conference (CITA)*, Beijing, China. December 2023.

- (b) *The 3rd Huairou Forum Biomedical Imaging: Future Technologies and Future Talents*, Beijing, China. November 2023.
- (c) *AI for Science Congress*, Beijing, China. August 2023.
4. “Computational imaging for exoplanet detection and characterization.”
 - (a) *Nanjing Institute of Astronomical Optics & Technology (NIAOT)*, CAS. October 2023.
 - (b) *Changchun Institute of Optics, Fine Mechanics and Physics (CIOMP)*, CAS. June 2023.
 - (c) *The 5th Young Scientist Forum of Planetary Science*, Hainan, China. March 2023.
 5. “Adaptive optics for high-contrast exoplanet direct imaging.”
 - (a) *The 6th Atmospheric Optics and Adaptive Optics Workshop*, Chengdu, China. October 2023.
 - (b) *Nanjing Institute of Astronomical Optics & Technology (NIAOT)*, CAS. October 2023.
 - (c) *Peking University, Aero&Astro Department*. May 2020.
 6. “Computational astronomical imaging: inference and sensing algorithms for future black hole observations.”
 - (a) *Shanghai Astronomical Observatory*, CAS. July 2023.
 - (b) *Chinese Conference on Pattern Recognition and Computer Vision (PRCV)*, Shenzhen, China. November 2022.
 - (c) *The 1st International Computational Imaging Conference (CITA)*, Shanghai, China. October 2022.
 - (d) *UCLA Warren Grundfest Memorial Lectures*. October 2021.
 - (e) *TU Delft Imaging Physics Department*. February 2021.
 7. “Computational scientific imaging: from astronomy to biomedicine.”
 - (a) *Tsinghua University Department of Astronomy & Department of Precision Instrument*. September 2022.
 - (b) *Peking University National Biomedical Imaging Center*. August 2021.
 8. “Efficient wavefront sensing and control with system identification and deformable mirror sensing command optimization.” *UCSB Direct Imaging Tech Webinar*. July 2020.
 9. “Adaptive optics with online system identification.” *Center for Adaptive Optics (CfAO) Fall Science Retreat*. Lake Arrowhead, CA. November 2019.

Teaching

Ph.D. Supervision	Weimin Bai - <i>Ph.D. in Biomedical Engineering</i> , 2023-present Enze Ye - <i>Ph.D. in Biomedical Engineering</i> , 2023-present Shaochi Ren - <i>Ph.D. in Biomedical Engineering</i> , 2023-present
Master Supervision	Yaqing Xu - <i>M.Eng. in Software Engineering</i> , 2022-present Zhiming Chang - <i>M.Eng. in Mechanical Engineering</i> , 2022-present
Instructor	<i>Adaptive Optics: Principles and Applications</i> , Peking University, Spring 2024 <i>Inverse Problems and Computational Imaging</i> , Peking University, Fall 2023
Guest Lecturer	<i>Artificial Intelligence for Scientific Computing</i> (by Bin Dong & Weinan E), Peking University, Spring 2023 <i>Artificial Intelligence for Precision Medicine</i> (by Congying Wu), Peking University, Summer 2022 <i>Computational Cameras</i> (by Katie Bouman), Caltech, Spring 2021 & 2022 <i>Data-Driven Algorithm Design</i> (by Yisong Yue), Caltech, Spring 2020
Teaching Assistant	<i>Automatic Control System</i> (by Clancy Rowley), Princeton, Fall 2018 <i>Introduction to Dynamics</i> (by N.Jeremy Kasdin), Princeton, Spring 2018 <i>Space Flight</i> (by N.Jeremy Kasdin), Princeton, Fall 2017 <i>Mathematics in Engineering</i> (by Howard Stone), Princeton, Fall 2016

Selected Research Grants

1. PI, NSFC, “Dynamic Microscopy Reconstruction with Implicit Neural Representations”, 2023-2026
2. PI, NBIC, “Multi-Conjugate Adaptive Optics (MCAO) for Microscopy”, 2023-2024
3. PI, Amazon AI4Science Fellowship, “Reliable Scientific Imaging with Accurate Noise Modeling and Uncertainty Quantification”, 2020-2022

Academic Service

Technical Committee	IEEE Computational Imaging, 2022-present
Technical Committee	CSOE (Chinese Society for Optical Engineering) Computational Imaging, 2023-present
Technical Committee	CSIG (China Society of Image and Graphics) 3D Vision, 2023-present
Technical Committee	CMBA (China Medicinal Biotechnology Association) Biomedical Imaging, 2023-present
Organizer	Biomedical and Astronomical Signal Processing (BASP) Frontiers, 2023
Organizer	Computational Cameras and Displays (CCD) Workshop, CVPR 2022 & 2023
Area Chair	International Conference on Acoustics Speech and Signal Processing (ICASSP), 2023
Poster/Demo Chair	International Conference on Computational Photography (ICCP), 2022

Reviewer	<i>Astronomy & Astrophysics</i> , <i>RAS Techniques and Instruments</i> , <i>Journal of Astronomical Telescopes, Instruments, and Systems</i> , <i>IEEE Transactions on Image Processing</i> , <i>IEEE/CAA Journal of Automatica Sinica</i> , <i>Optics Letters</i> , <i>ICCV</i> , <i>ECCV</i> , <i>CVPR</i> , <i>ICCP</i> , <i>ICASSP</i>
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